ADM - TD2

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Table des matières

Exercice 1

1. $m = \frac{26+23+14+33+24+18}{6} = 23$
2. $V(E) = \frac{1}{6} \sum_{k=1}^{6} (x_k - m)^2 = 36$
3. $$\sigma = \sqrt{V(E)} = \sqrt{36} = 6$$
4. On fait une ACP centrée réduite (normée)

$$V_1' = \frac{v_1 - \mu}{\sqrt{n} \cdot \sigma} = \frac{26 - 23}{\sqrt{6} * 6} = \frac{3}{6\sqrt{6}} = \frac{1}{2\sqrt{6}} \approx 0.204$$

5. Quand l'ACP est normée, la matrice de dispersion est la matrice de corrélation.

$$\mathrm{corr}(V_1, V_2) = \frac{\mathrm{cov}(V_1, V_2)}{\sigma_{V_1} \cdot \sigma_{V_2}}$$

$$\mathrm{cov}(i, j) = \frac{1}{n} \sum_{k=1}^{n} (X_{ki} - g_i)(X_{kj} - g_j)$$

$$\begin{aligned}
\mathrm{cov}(V_1, V_2) &= \frac{1}{n} \sum_{k=1}^{n} \left(X_{kv_1} - g_1\right)\left(X_{kv_2} - g_2\right) \\
&= \frac{1}{6}[(26 - 23)(36 - 33.667) + (23 - 23)(__) \\
&+(14 - 23)(25 - 33.667) + (33 - 23)(45 - 33.667) \\
&+(24 - 23)(34 - 33.667) + (18 - 23)(30 - 33.667)] \\
&\approx 36.16 \\
\mathrm{corr}(V_1, V_2) &= \frac{36.16}{6 \cdot 6.128} \\
&\approx 0.984
\end{aligned}$$

8. $$A_1 \cdot U_1 = \begin{pmatrix} 0.204 & 0.155 & 0.642 \end{pmatrix} \begin{pmatrix} 0.628 \\ 0.610 \\ 0.489 \end{pmatrix} \approx 0.533^2$$

9. On a $\psi_{jk} = \sqrt{\lambda_k} U_{jk}$

Ainsi $\psi_{11} = \sqrt{\lambda_1} U_{11} = \sqrt{2.4161} * 0.628 = 0.976$

10. $$\frac{\varphi_{11}^2}{\lambda_1} = \frac{0.533^2}{2.4161} = 0.118$$

11. $$\frac{\psi_{11}^2}{\lambda_1} = \frac{0.976^2}{2.4161} = 0.394$$

12. $$\cos^2(U_1, 1) = \frac{\varphi_{11}^2}{\sum_j \varphi_{1j}^2} = \frac{0.533^2}{0.533^2 + (-0.439)^2 + (-0.016)^2} \approx 0.595$$

13. $$\cos^2(V_1) = \frac{\psi_{11}^2}{\sum_j} \psi_{1j}^2 = \frac{0.976^2}{0.976^2 + 0.205^2 + 0.074^2} \approx 0.952$$